

Unit 18: The Mean Value Theorem

1. The idea, in plain words

Imagine you drive 150 miles in 3 hours. Your average speed for the whole trip is 50 mph. Now here's a question that sounds obvious once you hear it: at some exact instant during that drive, didn't your speedometer have to read exactly 50 mph? Maybe you were going faster at times and slower at others — but somewhere in between, your speed had to pass through 50 exactly, because that's what "average" means.

That everyday intuition is the entire Mean Value Theorem (MVT), formalized. It says: for a smooth, unbroken function over an interval $[a, b]$, there must be at least one point in between where the function's instantaneous rate of change ($f'(c)$) exactly equals its average rate of change over the whole interval.

Picture it graphically: draw the secant line connecting the two endpoints $(a, f(a))$ and $(b, f(b))$ — its slope is the average rate of change. The Mean Value Theorem guarantees that somewhere between a and b , the curve has a tangent line that is perfectly parallel to that secant line. No matter how wiggly the curve is in between, it can't avoid having at least one point where its tangent matches the secant's slope exactly.

A simpler special case worth knowing: Rolle's Theorem. If a function starts and ends at the same height ($f(a) = f(b)$), then the secant line connecting them is perfectly flat (slope 0). MVT then guarantees a point in between where the tangent line is also flat — meaning $f'(c) = 0$ somewhere in the middle. This makes sense: if you leave from a point and come back to the exact same height, you must have paused or turned around somewhere along the way.

What you need to be careful about: MVT only works if the function is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) . If either of those breaks down — a hole, a jump, a vertical asymptote, or a sharp corner anywhere in the interval — the theorem's guarantee no longer applies, and you can't assume such a point c exists.

2. Toolbox

Rolle's Theorem: if f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists at least one c in (a, b) such that

$$f'(c) = 0$$

Mean Value Theorem (MVT): if f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists at least one c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

The general procedure to find c :

1. Check that f is continuous on $[a, b]$ and differentiable on (a, b) (for most everyday functions like polynomials, this is automatic — but watch for domain issues with fractions, roots, or corners).
2. Compute the average rate of change: $\frac{f(b) - f(a)}{b - a}$.
3. Compute $f'(x)$.

4. Set $f'(x)$ equal to the average rate of change from Step 2, and solve for x .
5. Keep only the solution(s) that actually fall inside the open interval (a, b) — discard any that fall outside, or that land exactly on an endpoint.

3. Common mistakes

- Forgetting to check the continuity and differentiability hypotheses. If the function has a break, a hole, an asymptote, or a corner anywhere inside $[a, b]$, MVT simply doesn't apply — you can't use it at all in that case.
- Keeping a solution for c that falls outside the open interval (a, b) , including one that lands exactly on an endpoint. Only genuine interior solutions count.
- Expecting a unique answer. MVT only guarantees at least one value of c — sometimes the algebra produces more than one valid solution inside the interval, and all of them are correct.
- Mixing up Rolle's Theorem's extra condition. Rolle's Theorem is just MVT in the special case where $f(a) = f(b)$ — don't apply the " $f'(c) = 0$ " shortcut unless you've actually confirmed the endpoint heights are equal.
- Sign errors computing the average rate of change. Double-check the order of subtraction: $\frac{f(b) - f(a)}{b - a}$, not the other way around.

4. Problem Set

Warm-up

For each problem, verify MVT applies, then find the value(s) of c guaranteed by the theorem.

1. $f(x) = x^2 - 4x$ on $[0, 4]$
2. $f(x) = x^2 + 2x$ on $[-1, 3]$
3. $f(x) = x^3$ on $[0, 2]$
4. $f(x) = x^2 - 1$ on $[1, 3]$
5. $f(x) = x^2 - 4x + 3$ on $[1, 3]$
6. $f(x) = 2x^2 - 3x$ on $[0, 2]$

Standard

7. $f(x) = x^3 - 3x$ on $[-1, 2]$
8. $f(x) = \frac{1}{x}$ on $[1, 3]$
9. $f(x) = x^3 - x$ on $[-1, 1]$
10. $f(x) = \sqrt{x}$ on $[0, 4]$
11. $f(x) = \sin x$ on $[0, \pi]$
12. $f(x) = x^2 - 6x + 5$ on $[0, 6]$

Challenge

13. $f(x) = \frac{1}{x}$ on $[-1, 1]$. Explain why the Mean Value Theorem does not apply here.
14. $f(x) = |x|$ on $[-2, 2]$. Explain why the Mean Value Theorem does not apply here.
15. $f(x) = x^3 - 3x^2 + 2$ on $[0, 3]$. Find all value(s) of c guaranteed by MVT. (Be careful — the algebra may produce a solution that must be discarded.)

16. Suppose f is differentiable everywhere and $f'(x) \leq 2$ for every x . Suppose also $f(0) = 0$ and $f(3) = 10$. Use the Mean Value Theorem to show this is impossible — that no such function can exist.
17. A function f is continuous on $[1, 5]$ and differentiable on $(1, 5)$, with $f(1) = 3$ and $f(5) = 15$. Use the Mean Value Theorem to show that at some point c in $(1, 5)$, $f'(c) = 3$, and explain in words why the function's instantaneous rate of change can't stay strictly below 3 (or strictly above 3) across the whole interval.

Applied

18. A car travels 150 miles in 3 hours. Using the Mean Value Theorem, explain why the car's speedometer must have read exactly 50 mph at some instant during the trip.
19. A company's profit was \$200 at the start of the year ($t = 0$) and \$1400 at the end ($t = 12$ months). Using MVT, show there must have been some moment when the instantaneous rate of profit growth was exactly \$100 per month.
20. A runner completes a 10 km race in exactly 40 minutes. Using MVT, argue that there must have been some instant during the race when their speed was exactly 15 km/h.
21. A rocket's height (in feet) satisfies $h(0) = 0$ and $h(10) = 2000$, where t is measured in seconds. Find the average velocity over this interval, and use MVT to explain why the rocket's instantaneous velocity must equal this value at some moment during the flight.